

Nguyen Hung Minh Tan



CONTACT INFORMATION

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ACADEMIC APPOINTMENTS

National University of Singapore, Singapore 2023–present
• Assistant Professor (Presidential Young Professor, tenure-track), Department of Mathematics

University of California, Los Angeles, CA, USA 2020–present
• Post-doctoral Scholar in Mathematics
• Mentor: Professor Stanley J. Osher

EDUCATION

Rice University, Houston, Texas, USA

Ph.D. in Electrical and Computer Engineering, 2014–2020
• Advisor: Professor Richard G. Baraniuk
• Thesis: Momentum-based Methods for Training and Designing Deep Neural Networks

M.S. in Electrical and Computer Engineering, 2014–2016
• Advisor: Professor Richard G. Baraniuk
• Thesis: Bridging Theory and Practice in Deep Learning with the Deep Rendering Model

B.S. in Electrical and Computer Engineering, 2011–2014

RESEARCH INTERESTS

My research develops mathematical foundations for reliable, interpretable, scalable, and controllable AI. I use tools from optimization, geometry, dynamical systems, optimal transport, statistics, and algebra to study how foundation models represent information, how representations evolve through attention, state-space, and mixture-of-experts architectures, and how symmetries and conservation laws determine meaningful model behaviour.

My current focus is programmable control of foundation models through activation steering, feedback control, distributional and kernelized steering, and compositional control, with applications to trustworthy AI, AI for mathematical research and education, AI for science, and foundation-model adaptation.

WORKSHOP ORGANIZATION

Behavioral Control in Foundation Models via First-Principles Approaches, Special Session, Asilomar Conference on Signals, Systems, and Computers, 2026.

Mathematical Foundations and Methodologies for Artificial Intelligence and Data-Driven Scientific Computing at the Institute for Mathematical Sciences, NUS, 2026.

Integration of Deep Neural Models and Differential Equations at the International Conference on Learning Representations (ICLR), 2020.

Integration of Deep Learning Theories at the Conference on Neural Information Processing Systems (NeurIPS), 2018.

AWARDED GRANTS PI, Cross Faculty Grant at NUS, *Unlocking the Power of Probabilistic Computing with Agentic AI*, SGD 250,000, 2026 - 2027.

PI, NUS Grand Challenge - AI+X/Core AI, *Mitigating Hallucinations in Large Language Models*, SGD 110,000, 2025 - 2027.

PI, Faculty Research Committee (FRC) for Tier 1 funding, *A Primal-Dual Framework for Transformers: Efficiency, Privacy, and Robustness*, 2024 - 2027.

PI, Faculty Research Committee (FRC) for Tier 1 funding, *Statistical Frameworks for Transformers: Representability, Efficiency, and Robustness*, 2024 - 2027.

PI, Toyota Research Institute Grant, *A Unified Framework for Building Large-Scale Language Models in Developing Countries: From Model and System to Data*, \$100,000, November, 2022 - November, 2023.

Co-PI, Singapore-Israel Scientific Research Program, *Mitigating Security and Privacy Risks of Large Language Models*, SGD 250,000, October, 2023 - March, 2025.

HONORS AND
AWARDS

NUS Presidential Young Professorship, 2024-2027.

Computing Innovation Postdoctoral Fellowship (CIFellows), 2020–2023.

J. Tinsley Oden Visiting Fellowship, 2022

AWS Cloud Credits for Research, \$50,000, September, 2018.

NSF Graduate Research Fellowship, 2016–2020.

Ford Foundation Fellowship Honorable Mention, 2016.

Neuroengineering IGERT: From Cells to Systems Fellowship, 2015 – 2017.

Texas Instruments Fellowship, 2014–2016. (Awarded to top incoming graduate students in the ECE department at Rice University)

Louis J. Walsh Scholarship in Engineering, 2013–2014. (Awarded to top undergraduate students in the ECE department at Rice University)

Gold Scholar on Coca-Cola’s Community College Academic Team, 2011.

TEACHING

Spring 2025, 2026, DSA5207: Text Processing and Interpretation with Machine Learning, NUS.

Spring 2024, DSA5202: Advanced Topics in Machine Learning, NUS.

Fall 2023, 2024, 2025, DSA5105: Principles of Machine Learning, NUS.

Spring 2023 and Summer 2022, Math156: Machine Learning, UCLA.

INVITED SEMINAR
PRESENTATIONS

Dynamical Properties of Tokens in Large Language Models *Minisymposium on Optimization Foun-*

dations of Generative AI Models, SIAM Conference on Optimization, UK, 2026.

Steering Large Language Models: A Geometric and Control-Theoretic Approach. *Huawei Merlion Seminar, Singapore, 2026.*

Angular Steering: Behavior Control via Rotation in Activation Space. *Information Theory and Applications Workshop, USA, 2026.*

Equivariant Neural Functional Networks for Transformers. *Asian Mathematical Conference, Chiang Mai, 2025.*

Unveiling the Hidden Structure of Self-Attention via Kernel Principal Component Analysis. *EASIAM Minisymposium “Advances in data-driven modeling and scientific computation” in CSIAM Conference, China, 2024.*

Transformer Meets Nonparametric Kernel Regression. *ECE Departmental CSP Seminar at the University of Michigan, USA, 2024.*

Transformer with Fourier Integral Attentions. *Deep Learning for Sequence Modeling Minisymposium, the SIAM Conference on Computational Science and Engineering (CSE), 2023.*

A Primal-Dual Framework for Transformers and Neural Networks. *Special Session on “Mathematics of Machine Learning”, Canadian Mathematical Society Winter Meeting, 2022.*

A Statistical Treatment of the Attention Mechanism in Transformers. *Analytics Center of Excellence (ACOE), IQVIA, Italy, 2022.*

Principled Models for Machine Learning. *Math Machine Learning Seminar, the Max Planck Institute for Mathematics in the Sciences and UCLA, USA, 2022.*

Momentum-Based and Fast Multipole Methods for Designing Deep Learning Models. *Mathematical Foundation of Deep Learning with the Applications to PDE Minisymposium, the 4th Annual Meeting of the SIAM Texas-Louisiana Section, 2021. (Invited talk)*

Principled Models for Machine Learning. *Applied Math Colloquium at UCLA, USA, 2022.*

Momentum-based Methods for Training and Designing Deep Neural Networks. *VinAI Research, Vietnam, 2020.*

Neural Rendering Model: Rethinking Neural Networks from the Joint Generation and Prediction Perspective. *NeurIPS Workshop on Integration of Deep Learning Theories, 2018.*

ADVISING

Postdoctoral researchers at NUS: Dr. Vo Ngoc Thieu (Next: Assistant Professor, Department of Computer Science, University of Bath, UK).

Ph.D. students at NUS: Viet-Hoang Tran, Laziz U. Abdullaev, Shiqi Jiang, Tho Tran, Ryan Lee, Maurice Mok, Minh-Khoi Nguyen-Nhat, Dung V. Nguyen, Benoit Guerand.

Undergraduate students at NUS: Trung-Khang Tran.

Other undergraduate/master mentees: Wanjia Liu, Master student in the Computer Science Department at Rice University (Next: Google); Si Hui Dai, Undergraduate student in the Computing + Mathematical Sciences Department at Caltech (Next: Ph.D. at Princeton); Ethan Perez, Undergraduate student in the Computer Science Department at Rice University. (Next: Ph.D. at

NYU)

PROFESSIONAL
SERVICES

Conference Area Chairs

- International Conference on Machine Learning (ICML)
- Conference on Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS)

Journal reviewing

- Journal of Machine Learning Research
- Transactions on Pattern Analysis and Machine Intelligence
- Transactions on Machine Learning Research
- Machine Learning with Applications
- Information and Inference
- IEEE Journal on Selected Areas in Information Theory
- Statistical Applications in Genetics and Molecular Biology

Other service

- Summer Undergraduate Research Fellowship Program at Caltech, Pasadena, California, *Project Mentor*
- Machine Learning Lunch at Rice University, Houston, Texas, *Organizer*
- Deep Learning Meeting at Rice University, Houston, Texas, *Organizer*

JOURNAL
PUBLICATIONS

1. Duc Thach Son Vu, **Tan M. Nguyen**, Weiqing Ren. “Learning and predicting dynamics of compositional multiphase mixtures using Graph Neural Networks”. *Journal of Computational Physics*, 2025.
2. Bao Wang*, **Tan M. Nguyen***, Andrea L. Bertozzi, Richard G. Baraniuk, Stanley J. Osher. “Scheduled Restart Momentum for Accelerated Stochastic Gradient Descent”. *SIAM Journal on Imaging Sciences*, 2022.
3. Bao Wang, Hedi Xia, **Tan M. Nguyen**, Stanley J. Osher. “How Does Momentum Benefit Deep Neural Networks Architecture Design? A Few Case Studies”. *Research in the Mathematical Sciences*, 2022.
4. Yue Wang, Jianghao Shen, Ting-Kuei Hu, Pengfei Xu, **Tan M. Nguyen**, Richard Baraniuk, Zhangyang Wang, Yingyan Lin. “Dual Dynamic Inference: Enabling More Efficient, Adaptive, and Controllable Deep Inference”. *IEEE Journal of Selected Topics in Signal Processing*, 2020.

CONFERENCE
PUBLICATIONS

1. Viet-Hoang Tran*, Khanh Vinh Bui*, Tan Lai Ngoc, Nam Nguyen, Tuan Quang Dam**, **Tan M. Nguyen****. “Conservation Laws for Modern Neural Architectures”. *International Conference on Machine Learning (ICML)*, 2026 (*journal-quality, acceptance rate: 26.60%; Spotlight paper, top 2.2% of all submissions and top 8.44% of all accepted papers*) (new paper, 0 citation).
2. Tho H. Tran, Huu-Tuan Nguyen, Thien-Hai Nguyen, Nhat-Tri Ho, Viet-Hoang Tran, Tho Quan**, **Tan M. Nguyen****. “Geometric and Stochastic Analysis of Discontinuities in Sparse Mixture-of-Experts”. *International Conference on Machine Learning (ICML)*, 2026 (*journal-quality, acceptance rate: 26.60%; Spotlight paper, top 2.2% of all submissions and top 8.44% of all accepted papers*) (new paper, 0 citation).

3. Thai Khanh Nguyen*, Vo Ngoc Bich Uyen*, Thieu N. Vo**, Cuong Pham**, **Tan M. Nguyen****. “MuonSSM: Orthogonalizing State Space Models for Sequence Modeling”. *International Conference on Machine Learning (ICML), 2026 (journal-quality, acceptance rate: 26.60%; Spotlight paper, top 2.2% of all submissions and top 8.44% of all accepted papers)* (new paper, 0 citation).
4. Viet-Hoang Tran*, Khanh Vinh Bui*, Van-Hoan Trinh, Tan Lai Ngoc, **Tan M. Nguyen**. “Functional Equivalence in Attention: A Comprehensive Study with Applications to Linear Mode Connectivity”. *International Conference on Machine Learning (ICML), 2026 (journal-quality, acceptance rate: 26.60%)* (new paper, 0 citation).
5. Laziz U. Abdullaev*, Noelle Y.L. Wong*, Ryan Lee T.Z.*, Shiqi Jiang, Minh-Khoi Nguyen-Nhat, **Tan M. Nguyen**. “Concept Heterogeneity-aware Representation Steering”. *International Conference on Machine Learning (ICML), 2026 (journal-quality, acceptance rate: 26.60%)* (new paper, 0 citation).
6. Duy-Tung Pham**, Viet-Hoang Tran*, Thieu N. Vo, **Tan M. Nguyen**. “Mixed-Curvature Tree-Sliced Wasserstein Distance”. *International Conference on Learning Representations (ICLR), 2026 (journal-quality, acceptance rate: 27.40%)* (3 citations).
7. Viet-Hoang Tran*, Thanh Q. Tran*, Thanh Chu, Trung-Khang Tran, Duy-Tung Pham, Tam Le**, **Tan M. Nguyen****. “Revisiting Tree-Sliced Wasserstein Distance Through the Lens of the Fermat–Weber Problem”. *International Conference on Learning Representations (ICLR), 2026 (journal-quality, acceptance rate: 27.40%)* (3 citations).
8. Viet-Hoang Tran*, Thanh Q. Tran*, Thanh Chu, Duy-Tung Pham, Trung-Khang Tran, Tam Le**, **Tan M. Nguyen****. “Tree-sliced Sobolev IPM”. *International Conference on Learning Representations (ICLR), 2026 (journal-quality, acceptance rate: 27.40%)* (3 citations).
9. Viet-Hoang Tran*, An Nguyen*, Benoit Guerand, Thieu N. Vo, **Tan M. Nguyen**. “Quasi-Equivariant Metanetworks”. *International Conference on Learning Representations (ICLR), 2026 (journal-quality, acceptance rate: 27.40%)* (0 citations).
10. Dung V. Nguyen*, Anh T. Nguyen*, Minh H. Nguyen, Luc Q. Nguyen, Shiqi Jiang, Ethan Fetaya, Linh Duy Tran**, Gal Chechik**, **Tan M. Nguyen****. “Expert Merging in Sparse Mixture of Experts with Nash Bargaining”. *International Conference on Learning Representations (ICLR), 2026 (journal-quality, acceptance rate: 27.40%)* (3 citations).
11. Dung V. Nguyen*, Nhi Y. Pham*, Hieu M. Vu*, Lei Zhang**, **Tan M. Nguyen****. “Activation Steering with a Feedback Controller”. *International Conference on Learning Representations (ICLR), 2026 (journal-quality, acceptance rate: 27.40%)* (2 citations).
12. Viet-Hoang Tran*, Van-Hoan Trinh*, Khanh Vinh Bui*, **Tan M. Nguyen**. “On Linear Mode Connectivity of Mixture-of-Experts Architectures”. *Conference on Neural Information Processing Systems (NeurIPS), 2025 (journal-quality, acceptance rate: 24.52%; Oral paper, top 0.36% of all submissions and top 1.46% of all accepted papers)* (6 citations).
13. Hieu M. Vu, **Tan M. Nguyen**. “Angular Steering: Behavior Control via Rotation in Activation Space”. *Conference on Neural Information Processing Systems (NeurIPS), 2025 (journal-quality, acceptance rate: 24.52%; Spotlight paper, top 3.19% of all submissions and top 13.01% of all accepted papers)* (16 citations).
14. Duy-Tung Pham, An Nguyen The, Viet-Hoang Tran, Nhan-Phu Chung, Xin T. Tong, **Tan M. Nguyen****, Thieu N. Vo**. “Dynamical Properties of Tokens in Self-Attention and Effects of Positional Encoding”. *Conference on Neural Information Processing Systems (NeurIPS), 2025 (journal-quality, acceptance rate: 24.52%)* (0 citations).
15. Viet-Hoang Tran, Thanh Tran, Thanh Chu, Tam Le**, **Tan M. Nguyen****. “Tree-Sliced Entropy Partial Transport”. *Conference on Neural Information Processing Systems (NeurIPS), 2025 (journal-quality, acceptance rate: 24.52%)* (5 citations).

16. Rachel S.Y. Teo*, Laziz U. Abdullaev*, **Tan M. Nguyen**. “The Blessing and Curse of Dimensionality in Safety Alignment”. *Conference on Language Modeling (COLM)*, 2025 (acceptance rate: 32.00%) (4 citations).
17. Thieu N. Vo*, Hoang V. Tran*, Tho H. Tran*, An T. Nguyen, Thanh Tran, Minh-Khoi Nguyen-Nhat, Duy-Tung Pham, **Tan M. Nguyen**. “Equivariant Polynomial Functional Networks”. *International Conference on Machine Learning (ICML)*, 2025 (journal-quality, acceptance rate: 26.90%) (0 citations).
18. Hoang V. Tran*, Huyen Trang Pham*, Tho H. Tran, Minh-Khoi Nguyen-Nhat, Thanh Chu, Tam Le**, **Tan M. Nguyen****. “Tree-Sliced Wasserstein Distance: A Geometric Perspective”. *International Conference on Machine Learning (ICML)*, 2025 (journal-quality, acceptance rate: 26.90%) (11 citations).
19. Thanh Tran*, Hoang V. Tran*, Thanh Chu, Huyen Trang Pham, Laurent El Ghaoui, Tam Le**, **Tan M. Nguyen****. “Tree-Sliced Wasserstein Distance with Nonlinear Projection”. *International Conference on Machine Learning (ICML)*, 2025 (journal-quality, acceptance rate: 26.90%) (14 citations).
20. Ngoc-Quan Pham*, Tuan Truong*, Quyen Tran, **Tan M. Nguyen**, Dinh Phung, Trung Le. “Promoting Ensemble Diversity with Interactive Bayesian Distributional Robustness for Fine-tuning Foundation Models”. *International Conference on Machine Learning (ICML)*, 2025 (journal-quality, acceptance rate: 26.90%) (3 citations).
21. Rachel S.Y. Teo, **Tan M. Nguyen**. “MoLEx: Mixture of Layer Experts for Fine-tuning with Sparse Upcycling”. *International Conference on Learning Representations (ICLR)*, 2025 (journal-quality, acceptance rate: 32.08%) (3 citations).
22. Laziz U. Abdullaev, **Tan M. Nguyen**. “Transformer Meets Twicing: Harnessing Unattended Residual Information”. *International Conference on Learning Representations (ICLR)*, 2025 (journal-quality, acceptance rate: 32.08%) (7 citations).
23. Stefan K. Nielsen*, Rachel S.Y. Teo*, Laziz U. Abdullaev, **Tan M. Nguyen**. “Tight Clusters Make Specialized Experts”. *International Conference on Learning Representations (ICLR)*, 2025 (journal-quality, acceptance rate: 32.08%) (8 citations).
24. Thieu N. Vo, Duy-Tung Pham, Xin T. Tong, **Tan M. Nguyen**. “Demystifying the Token Dynamics of Deep Selective State Space Models”. *International Conference on Learning Representations (ICLR)*, 2025 (journal-quality, acceptance rate: 32.08%; **Spotlight paper**, top 5.10% of all submissions and top 10.26% of all accepted papers) (4 citations).
25. Viet Dung Nguyen, Minh Nguyen Hoang, Rachel S.Y. Teo, Luc Nguyen, **Tan M. Nguyen****, Linh Duy Tran**. “CAMEx: Curvature-aware Merging of Experts”. *International Conference on Learning Representations (ICLR)*, 2025 (journal-quality, acceptance rate: 32.08%) (3 citations).
26. Hoang V. Tran*, Thieu N. Vo*, An T. Nguyen, Tho H. Tran, Minh-Khoi Nguyen-Nhat, Thanh Tran, Duy-Tung Pham, **Tan M. Nguyen**. “Equivariant Neural Functional Networks for Transformers”. *International Conference on Learning Representations (ICLR)*, 2025 (journal-quality, acceptance rate: 32.08%) (1 citations).
27. Hoang V. Tran*, Minh-Khoi Nguyen-Nhat*, Huyen Trang Pham, Thanh Chu, Tam Le**, **Tan M. Nguyen****. “Distance-Based Tree-Sliced Wasserstein Distance”. *International Conference on Learning Representations (ICLR)*, 2025 (journal-quality, acceptance rate: 32.08%) (17 citations).
28. Hoang V. Tran*, Thanh Chu*, Minh-Khoi Nguyen-Nhat, Huyen Trang Pham, Tam Le**, **Tan M. Nguyen****. “Spherical Tree-Sliced Wasserstein Distance”. *International Conference on Learning Representations (ICLR)*, 2025 (journal-quality, acceptance rate: 32.08%) (17 citations).

29. Rachel S.Y. Teo, **Tan M. Nguyen**. “MomentumSMoE: Integrating Momentum into Sparse Mixture of Experts”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2024 (journal-quality, acceptance rate: 25.8%) (5 citations).
30. Rachel S.Y. Teo, **Tan M. Nguyen**. “Unveiling the Hidden Structure of Self-Attention via Kernel Principal Component Analysis”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2024 (journal-quality, acceptance rate: 25.8%) (14 citations).
31. Stefan Nielsen*, Laziz Abdullaev*, Rachel S.Y. Teo, **Tan M. Nguyen**. “Elliptical Attention”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2024 (journal-quality, acceptance rate: 25.8%) (14 citations).
32. Hoang V. Tran*, Thieu Vo*, Tho Tran Huu, An Nguyen The, **Tan M. Nguyen**. “Monomial Matrix Group Equivariant Neural Functional Networks”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2024 (journal-quality, acceptance rate: 25.8%) (13 citations).
33. Tam Nguyen, Cesar A. Uribe, **Tan M. Nguyen****, Richard Baraniuk**. “PIDformer: Transformer Meets Control Theory”. *International Conference on Machine Learning (ICML)*, 2024 (journal-quality, acceptance rate: 27.5%) (15 citations).
34. Hien Dang, Tho Tran Huu, **Tan M. Nguyen****, Nhat Ho**. “Neural Collapse for Cross-entropy Class-Imbalanced Learning with Unconstrained ReLU Features Model”. *International Conference on Machine Learning (ICML)*, 2024 (journal-quality, acceptance rate: 27.5%) (24 citations).
35. Long Minh Bui, Tho Tran Huu, Duy Dinh, **Tan M. Nguyen****, Trong Nghia Hoang**. “Revisiting Kernel Attention with Correlated Gaussian Process Representation”. *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2024 (journal-quality, acceptance rate: 27.0%) (12 citations).
36. Hien Dang, Tho Tran Huu, **Tan M. Nguyen****, Nhat Ho**. “Beyond Vanilla Variational Autoencoders: Detecting Posterior Collapse in Conditional and Hierarchical Variational Autoencoders”. *International Conference on Learning Representations (ICLR)*, 2024 (journal-quality, acceptance rate: 30.8%) (12 citations).
37. Tuan Nguyen, Hirotada Honda, Takashi Sano, Vinh Nguyen**, Shugo Nakamura**, **Tan M. Nguyen****. “From Coupled Oscillators to Graph Neural Networks: Reducing Over-smoothing via a Kuramoto Model-based Approach”. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024 (journal-quality, acceptance rate: 27.6%) (23 citations).
38. Tam Nguyen, **Tan M. Nguyen**, Richard G. Baraniuk. “Mitigating Over-smoothing in Transformers via Regularized Nonlocal Functionals”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2023 (journal-quality, acceptance rate: 26.1%) (40 citations).
39. Xing Han*, Tongzheng Ren*, **Tan M. Nguyen***, Khai Nguyen, Joydeep Ghosh, Nhat Ho. “Designing Robust Transformers using Robust Kernel Density Estimation”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2023 (journal-quality, acceptance rate: 26.1%) (17 citations).
40. Khang Nguyen, Hieu Nong, Vinh Nguyen, Nhat Ho, Stanley J. Osher, **Tan M. Nguyen**. “Revisiting Over-smoothing and Over-squashing using Ollivier-Ricci Curvature”. *International Conference on Machine Learning (ICML)*, 2023 (journal-quality, acceptance rate: 27.9%) (182 citations).
41. Hien Dang, Tho Tran Huu, Stanley Osher, Hung Tran-The, Nhat Ho, **Tan M. Nguyen**. “Neural Collapse in Deep Linear Networks: From Balanced to Imbalanced Data”. *International Conference on Machine Learning (ICML)*, 2023 (journal-quality, acceptance rate: 27.9%) (52 citations).
42. **Tan M. Nguyen***, Tam Nguyen*, Nhat Ho, Andrea Bertozzi, Richard G. Baraniuk, Stanley J. Osher. “A Primal-Dual Framework for Transformers and Neural Networks”. *International Conference on Learning Representations (ICLR)*, 2023 (journal-quality, acceptance rate:

- 32.0%; **Spotlight paper**, top 5.68% of all submissions and top 17.79% of all accepted papers) (28 citations).
43. Khai Nguyen, Tongzheng Ren, Huy Nguyen, Litu Rout, **Tan M. Nguyen**, Nhat Ho. “Hierarchical Sliced Wasserstein Distance”. *International Conference on Learning Representations (ICLR)*, 2023 (journal-quality, acceptance rate: 32.0%) (42 citations).
 44. Khang Nguyen, Hieu Nong, Khuong Nguyen, **Tan M. Nguyen****, Vinh Nguyen**. “Deep-GRAND: Deep Graph Neural Diffusion”. *Asilomar Conference*, 2023 (0 citation).
 45. **Tan M. Nguyen***, Tam Nguyen*, Long Bui, Hai Do, Dung Le, Hung Tran-The, Khuong Nguyen, Richard G. Baraniuk, Nhat Ho, Stanley J. Osher. “A Probabilistic Framework for Pruning Transformers via a Finite Admixture of Keys”. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP, notable-top-3%)*, 2023 (7 citations).
 46. **Tan M. Nguyen***, Minh Pham*, Tam Nguyen, Khai Nguyen, Stanley J. Osher, Nhat Ho. “FourierFormer: Transformer Meets Generalized Fourier Integral Theorem”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2022 (journal-quality, acceptance rate: 25.6%) (65 citations).
 47. **Tan M. Nguyen***, Tam Nguyen*, Hai Do, Khai Nguyen, Vishwanath Saragadam, Minh Pham, Khuong Nguyen, Nhat Ho, Stanley J. Osher. “Improving Transformer with an Admixture of Attention Heads”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2022 (journal-quality, acceptance rate: 25.6%) (40 citations).
 48. Nghia Nguyen*, **Tan M. Nguyen***, Huyen Vo, Stanley J. Osher, Thieu Vo. “Improving Neural Ordinary Differential Equations with Nesterov’s Accelerated Gradient Method”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2022 (journal-quality, acceptance rate: 25.6%) (27 citations).
 49. Tam Nguyen*, **Tan M. Nguyen***, Dung Le, Khuong Nguyen, Anh Tran, Richard G. Baraniuk, Nhat Ho, Stanley J. Osher. “Improving Transformers with Probabilistic Attention Keys”. *International Conference on Machine Learning (ICML)*, 2022 (journal-quality, acceptance rate: 21.9%) (43 citations).
 50. Matthew Thorpe*, **Tan M. Nguyen***, Hedi Xia*, Thomas Strohmer, Andrea Bertozzi, Stanley J. Osher, Bao Wang. “GRAND++: Graph Neural Diffusion with a Source Term”. *International Conference on Learning Representations (ICLR)*, 2022 (journal-quality, acceptance rate: 32.9%) (144 citations).
 51. **Tan M. Nguyen**, Richard G. Baraniuk, Mike Kirby, Stanley J. Osher, Bao Wang. “Momentum Transformer: Closing the Performance Gap Between Self-attention and Its Linearization”. *Mathematical and Scientific Machine Learning (MSML)*, 2022 (10 citations).
 52. **Tan M. Nguyen**, Animesh Garg, Richard G Baraniuk, Anima Anandkumar. “InfoCNF: An Efficient Conditional Continuous Normalizing Flow with Adaptive Solvers”. *Asilomar Conference*, 2022 (15 citations).
 53. **Tan M. Nguyen**, Vai Suliafu, Stanley J. Osher, Long Chen, Bao Wang. “FMMformer: Efficient and Flexible Transformer via Decomposed Near-field and Far-field Attention”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2021 (journal-quality, acceptance rate: 25.7%) (39 citations).
 54. Hedi Xia, Vai Suliafu, Hangjie Ji, **Tan M. Nguyen**, Andrea Bertozzi, Stanley J. Osher, Bao Wang. “Heavy Ball Neural Ordinary Differential Equations”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2021 (journal-quality, acceptance rate: 25.7%) (92 citations).
 55. **Tan M. Nguyen**, Richard G. Baraniuk, Andrea Bertozzi, Stanley J. Osher, Bao Wang. “MomentumRNN: Integrating Momentum into Recurrent Neural Networks”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2020 (journal-quality, acceptance rate: 20.1%) (45 citations).

56. Yujia Huang, James Gornet, Sihui Dai, Zhiding Yu, **Tan M. Nguyen**, Doris Tsao, Anima Anandkumar. “Neural Networks with Recurrent Generative Feedback”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2020 (*journal-quality, acceptance rate: 20.1%*) (69 citations).
57. **Tan M. Nguyen***, Nhat Ho*, Ankit B. Patel, Anima Anandkumar, Michael I. Jordan, Richard G. Baraniuk. “Neural Rendering Model: Joint Generation and Prediction for Semi-Supervised Learning”. *Deep Math Conference (DeepMath)*, 2019. (Oral presentation) (8 citations)
58. Ankit B Patel, **Tan M. Nguyen**, Richard Baraniuk. “A Probabilistic Framework for Deep Learning”. *Conference on Neural Information Processing Systems (NeurIPS)*, 2016 (*journal-quality, acceptance rate: 23.6%*) (98 citations).

WORKSHOP PAPERS

1. Gavin D. Portwood, Peetak P. Mitra, Mateus Dias Ribeiro, **Tan M. Nguyen**, Balasubramanya T. Nadiga, Juan A. Saenz, Michael Chertkov, Animesh Garg, Anima Anandkumar, Andreas Dengel, Richard G. Baraniuk, David P. Schmidt. “Turbulence Forecasting via Neural ODE”. *NeurIPS Workshop on Machine Learning and the Physical Sciences*, 2019.

REFERENCES

Professor Zuowei Shen

Distinguished Professor of Mathematics
 Director of the Institute for Mathematical Sciences
 National University of Singapore, Singapore
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Professor Stanley J. Osher

Professor of Mathematics, Computer Science, Electrical Engineering, and Chemical and Biomolecular Engineering
 Director of Special Projects, Institute for Pure and Applied Mathematics (IPAM)
 University of California, Los Angeles
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Professor Andrea L. Bertozzi

Distinguished Professor of Mathematics and Mechanical and Aerospace Engineering
 Betsy Wood Knapp Chair for Innovation and Creativity
 Director of Applied Mathematics
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Professor Richard G. Baraniuk

C. Sidney Burris Professor of Electrical and Computer Engineering
 Founder & Director, OpenStax
 Rice University, Houston, Texas
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